Draft version April 24, 2026
Typeset using LaTeX **modern** style in AASTeX631

Frizzle: Combining spectra or images by forward modeling

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ABSTRACT

When there are many observations of an astronomical source—many images with different dithers, or many spectra taken at different barycentric velocities—it is standard practice to shift and stack the data, to (for example) make a high signal-to-noise average image or mean spectrum. Bound-saturating measurements are made by manipulating a likelihood function, where the data are treated as fixed, and model parameters are modified to fit the data. Traditional shifting and stacking of data can be converted into a model-fitting procedure, such that the data are not modified, and yet the output is the shift-adjusted mean. The key component of this conversion is a spectral model that is completely flexible but also a continuous function of wavelength (or position in the case of imaging) that can represent any signal being measured by the device after any reasonable translation (or rotation or field distortion). The benefits of a modeling approach are myriad: The sacred data never are modified. Noise maps, data gaps, and bad-data masks don't require interpolation. The output can take the form of an image or spectrum evaluated on a pixel grid, as is traditional. In addition to shifts, the model **could be extended to** account for line-spread or point-spread function variations, world-coordinate-system variations, and calibration or normalization variations. The noise in the output becomes uncorrelated across neighboring pixels as the shifts deliver good coverage in some sense. We demonstrate the method with a small data example and provide open-source code for re-use.

Keywords: data analysis — image processing — spectroscopy — statistics

1. INTRODUCTION

It would be wrong to say that the practice of astronomy is, basically, *staring at the sky*. That said, staring at the sky is a big part of what we do. And the way we do it, usually, is this: We take many individual images of our target with exposure times that are long enough to make read noise irrelevant (if possible) while at the same time short enough to avoid saturation (of what we care about). We then shift and

* Authorship order was resolved by a game of *Chicken* that lasted *two years*: the first author would be whoever cracked and responded to the (perfectly reasonable and very helpful) referee report.

average (maybe median?) those images to get the highest possible signal-to-noise on our target. This, fundamentally, is the technology underlying the *Hubble Deep Field* (Williams et al. 1996), the *APOGEE* spectroscopic survey (Majewski et al. 2017), and the future Rubin Observatory *LSST* (Ivezić et al. 2019), among countless other projects. This technology is encoded in tools like `Drizzle` (Fruchter & Hook 2002).

These traditional practices are reliable and fast. However, they will—in some cases—sacrifice some information. Minimum-variance unbiased estimators require care in their construction; they are generally maximum-likelihood estimators (Voinov & Nikulin 2012). The deep, high signal-to-noise image or spectrum we seek is a statistical estimate of some kind of "true" or high signal-to-noise image or spectrum, given the noisy data of the individual exposures. That is, a combined image is a kind of *prediction*: It is the prediction of what *would have been observed* with a much longer exposure with a much better device. In measurement theory, the practice is to combine noisy data into a measurement by building and optimizing a likelihood. Of course we don't always think of a combined (or mosaiced) image of the sky or of an average spectrum of a star as the result of applying an *estimator*, but it is. It is just a simultaneous estimate of a lot of different components—a lot of different pixels—all the pixels in the combined image or spectrum.

We are in a period in astrophysics in which exceedingly precise measurements are expected, in both imaging and spectroscopy projects. In spectroscopy, in particular, high signal-to-noise spectral measurements are expected to obtain surface-abundance measurements with precisions of a few percent, and radial-velocity measurements with precisions of up to one ten-thousandth of a *pixel* (in terms of the Doppler shift). Spectral combination methods that distort absorption (or emission) lines will translate into biases, errors, or increased variance in these measurements. Spectral combination methods that correlate the noise in the output combined spectrum will translate into increased variance, and necessitate more sophisticated downstream measurement techniques.

One of the great things about the standard practice—the simple process of shifting and stacking the data—is that this procedure makes almost no assumptions about what the output, combined image or spectrum will look like. It isn't really a "model" for the data; it is more like a "summary" of the data **in that the output pixels are summary statistics of the input data**. When we switch to a statistical-inference framework, we have to choose some form or basis for expressing the combined image. In what follows we use a maximally parameterized model, in the sense that we choose an expression for the functions or function space that has as many parameters as the output image has pixels. That means that the approach we take does not get its power from the choice of parameterized functions for the output: Indeed it has just as much freedom as the shifting-and-stacking standard practice. The forward-modeling we recommend gets its power from the fact that it generates the input data directly and without manipulation, by means of a justifiable likelihood function.

A boundary condition of the work presented here is that the functional requirements of anything we provide must be the same as with standard practice: That is, we consider here only procedures that start with individual, single-epoch observa-

tions as inputs, along with noise estimates and bad-pixel masks; and we consider only procedures that end with an output "combined image" or "combined spectrum" that is a set of intensity or flux values on a pixel grid, along with uncertainty estimates and metadata flags. That is, the only difference between standard practice and the forward model that we advocate is purely in the nature of the estimator. Unlike the shift-and-coadd approach, the estimator we will deliver will be insensitive to under- or poorly-sampled data, which is not uncommon when astronomical seeing accidentally gets excellent, or in spectrographs that have limited detector pixels (for example, Wilson et al. 2019). As we will show with artificial data, the estimator we propose yields a faithful estimate of the *true* spectrum, and usually with good noise properties. Both of these results make our results good for subsequent precision measurements (e.g., radial velocities, chemical abundances).

The approach we propose differs from interpolation-and-averaging in a subtle way. Interpolation could be described as computing new values at arbitrary positions based on an existing set of known data points. Instead, here we propose to construct a continuous spectral model for the data, such that we can evaluate that model on either the input (raw-data) pixel grid or else on the output (combined spectrum) pixel grid. The *output product* is the same form as what one would expect from an interpolation-and-averaging procedure (values at a set of pixel positions), but the *procedure* differs from interpolation. When interpolating, the basis functions are explicitly constructed from the input-data pixel grid, whereas here we construct a forward model that is maximally consistent with the data, and can be evaluated at any pixel locations.

The ideas here aren't novel; many projects have avoided data interpolation previously. For one example, the *COMBO-17* survey (Wolf et al. 2003) took many images of the sky in many bands, and did not co-add the data prior to making measurements, in part to avoid mixing different point-spread functions and pixel samplings. For another, some projects have sought to co-add background-dominated images in a near optimal way, but presume that all observations are already registered and re-sampled to the same grid (Zackay & Ofek 2017). For another, the *DESI* survey ensures that all spectra are extracted on the same wavelength grid, to avoid any interpolation or resampling (Guy et al. 2023). What we propose below is actually a sub-problem of the problem solved by the *wobble* method for determining stellar radial velocities (Bedell et al. 2019); *wobble* is more general because it also includes a tellurics model and also tellurics and stellar variability models, plus it learns (it isn't told) the epoch-to-epoch spectral shifts. There is a heritage for what's presented here in the "optimal extraction" literature (for example, Hewett et al. 1985, Kelson 2003, Zechmeister et al. 2014, Bolton & Schlegel 2010): Optimal extraction describes a process of extracting a one-dimensional spectral signal from a two-dimensional spectrograph image when that extraction proceeds by forward modeling the two-dimensional data.[1] One amusing point, which is out of scope here, is that if the data that are being combined are extracted one-dimensional spectra, and those are, in turn, optimal extractions, then there is almost certainly an even better method that moves this spectral combination

[1] Forward modeling refers to predicting data given some model parameters. Inverse modeling describes the process of inferring some model parameters given some data.

method directly into the two-dimensional data; that is, there ought to be a method that obviates one-dimensional spectral extraction entirely. We hope for a future in which one-dimensional spectra are never extracted from two-dimensional images; in this future the two-dimensional data will be the sacred data, not the extractions.

In real applications, the spectra we extract are often of variable sources, or taken at different position angles on extended objects, such that we don't expect each epoch to be identical. Similarly, the *LSST* data are being taken the way they are because the sky (and point-spread function) will vary. Hence, in assuming that we only want the time-invariant mean of our data, we are considering here the very easiest problem in a range of possible problems. However, we note that all standard shift-and-coadd methods make similar assumptions implicitly. Part of our contribution in this *Article* is to make implicit assumptions explicit; we do this in Section 2.

Finally it is worthy of remark that, in some conceptual sense, there is no reason to *ever* combine the individual spectroscopic exposures at all, for any reason, except possibly visualization and good vibes. No information is gained by performing a combination. Any line or feature that becomes visible under spectral combination must—at least in principle—be measurable in the uncombined individual spectra if those individual spectra are analyzed appropriately simultaneously. That was the insight and motivation for the unusual *COMBO-17* (Wolf et al. 2003) approach in which the individual exposures were never co-added or combined; it is also the underlying idea behind work we have done on the proper motions and parallaxes of very faint sources (Lang et al. 2009). Averaging the data doesn't ever create any new information. It can be satisfying though; it meets astronomical community expectations; and it might provide insights or inspire discoveries.

2. CONCEPTS AND ASSUMPTIONS

What we do here can apply to **many kinds of** astronomical signals. **In addition to the direct applications of images and spectroscopy from ultraviolet to the infrared, Frizzle could represent an unseen image plane where complex visibilities are partially observed through interferometry (e.g., Hilder et al. 2025). However, here we are going to specialize to multi-epoch astronomical spectra for specificity.** We are going to make many other assumptions too, all of which are discussed in this Section.

Multi-epoch spectra with shifts — We will assume that there are $N > 1$ observations (epochs) of the same star (or maybe of different stars or quasars, say, that are assumed to be more-or-less identical, intrinsically, but let's think of a single star for now). Each of these observations i (with $1 \leq i \leq \mathbf{N}$) produces a one-dimensional observed (noisy) image $\mathbf{y}_i$. The $\mathbf{y}_i$ can be thought of as M_i-vectors in what follows, **with scalar components $y_{i,k}$ for $1 \leq k \leq M_i$,** where M_i is the number of pixels in spectrum i. **Each epoch can have a different length vector of pixel values.** The different observations are made at different relative velocities (star minus spectrograph) such that the images are shifted by a different shift $\Delta\mathbf{x}_i$ relative to the detector or extracted wavelength grid.

Shift operators — Because the shifts might be non-linear, the shifts $\Delta\mathbf{x}_i$ won't be pure numbers but something more like shift operators. That is, if there were a true spectrum $\tilde{y}$, each individual noisy image $\mathbf{y}_i$ would be related to that true spectrum by a shift operator acting on that true spectrum, plus noise. We will assume here that (from some external information) we know these shift operators very accurately. This is often not true, of course; we will return to this point in Section 5 below.

Wavelength calibration — We will assume that, for every M_i-pixel image $\mathbf{y}_i$ we also have a M_i-vector (or list) $\mathbf{x}_i$ of pixel positions, such that each element $x_{i,k}$ of $\mathbf{x}_i$ gives an accurate and precise value for the wavelength (or position in the spectrograph, in some settings) for each pixel of the image $\mathbf{y}_i$. That is, the device is well calibrated, and we know all the housekeeping data we need for image $\mathbf{y}_i$.

If the data being combined were images rather than spectra, the relevant calibration would be astrometric calibration; to wit, that each pixel has a known position on the sky.

Noise model — Each observation is noisy. We will assume that the noise has good properties, and especially that it is additive, (nearly) zero-mean and (nearly) Gaussian in form. We will *not* assume that you know the variance of the noise accurately, but you can do better with this method if you do know the noise variances accurately for each pixel. In what follows, each spectrum $\mathbf{y}_i$ is given an associated (but possibly inaccurate) noise-variance tensor C_i. If $\mathbf{y}_i$ is an M_i-vector, then C_i must be a nonnegative-definite $M_i \times M_i$ 2-tensor. In this description, there is no need to assume that the different pixels of the spectrum received independent noise, but we do assume that each epoch i has an independent noise draw.

Bad-pixel masks — Occasionally (or frequently) some of the pixels of a spectrum $\mathbf{y}_i$ might be bad—affected by cosmic rays or electronics issues—such that there is a bad-pixel mask $\mathbf{b}_i$ **with components** $b_{i,k}$ associated with each spectrum $\mathbf{y}_i$. This mask has value 1 in the locations in all good pixels, and 0 in the locations of all bad pixels. There are many different ways in which a pixel can be bad; in what follows the assumption is that the relevant judgements have been made more-or-less correctly.

Data gaps — There is no assumption here that all the $\mathbf{y}_i$ have the same number of pixels M_i, nor that the pixel grid is in any sense uniform or complete. That is, there can be gaps and spaces in the wavelength coverage of the spectra, as there are at chip gaps and where the detector has bad columns.

Pixel-convolved line-spread function — The spectrograph has something like a point-spread function or line-spread function, which, in the case of spectroscopy, sets the shape of an unresolved spectral line. In general it depends on the slit or fiber width, the spectrograph resolution and optics, instrument focus, and the properties of the detector. In what follows we will not build a model of any of this. Instead we will assume that there is a relatively constant pixel-convolved line-spread function (PCLSF), constant both in time, and constant with respect to the shifts $\Delta\mathbf{x}_i$ we see in our data set. We consider only the PCLSF, because then the sampling by the detector pixels does not require any additional convolution. That is, when the PCLSF is used

to model a spectrum, the projection onto the pixels is just a sampling of the function at the pixel centers, and does not involve any convolution over the face of the pixel. All that said, we won't explicitly model or use the PCLSF in what follows; we will just assume that our continuous spectral model is a model for the pixel-convolved, finite-resolution spectrum; we won't be doing any convolutions in comparing models and data.

In many cases the LSF or PSF is actually substantially different between exposures. This leads to spectral (or image) variations, and violate the assumptions of this model (and also the implicit assumptions underlying standard practice). We won't address this kind of variability here, except for some comments in Section 5: The method presented here **could be modified to account for differential LSF or PSF variations by having the model represent a 'highest resolution' spectrum or image, and applying different LSF or PSF kernels to each observation before comparing to data, but this extension is outside the scope of this paper.**

Continuous spectral model — In what follows we are going to avoid interpolating the data. We can avoid it by having a spectral *model* instead that can be arbitrarily and losslessly interpolated. That means that the spectral model must be a representation of a continuous one-dimensional function. If we were working on two-dimensional images, the model would have to be a representation of a continous two-dimensional function.

Linear basis or mixture models — The simplest kind of model for a continuous function is a sum or mixture or linear combination of continuous basis functions. These could be sines and cosines for a Fourier series model. They could be spline (or sinc or Lanczos) cardinal basis functions. They could be wavelets. The point is that if we use a linear basis, and assume that noises are Gaussian, we will get closed-form expressions for the data combination**, even if the basis is not orthogonal**. Thus we will make use of a linear basis.

Band limit — We are not assuming, in what follows, that the result we are looking for will be band-limited; we say that a signal is *band-limited* if it has no frequency content above some highest possible cutoff frequency. There is a weird sense in which data from a spectrograph both is, and is not, band limited. It is band limited in that the spectrograph is finite in resolution, such that no real spectroscopic signal can show features narrower than the PCLSF. It is not band-limited in that the photon noise in the device is independent (or close to independent) from pixel to pixel, such that the noise contribution to each observation $\mathbf{y}_i$ has support at all spatial frequencies.

One interesting question in what follows is whether the forward model we build for the data should itself be restricted to the band limit defined by the spectrograph resolution, or whether it should permit higher frequencies. It might surprise the reader to learn that we prefer to let the model capture higher frequencies, even when we think they shouldn't be there for any reasons other than noise, given the hardware. The reason we provide our model the flexibility to go to higher frequencies is that we want

our combined spectrum to have the capacity to capture or represent the pixel-to-pixel noise in the estimate; this pixel-to-pixel noise obeys no band limit.

Critical sampling — We are not assuming, in what follows, that the data are critically sampled. Because of considerations related to the Nyquist frequency on a uniform grid, it is valuable for spectra to be represented with a pixel spacing that is smaller than the width of the PCLSF. Specifically, it is valuable for there to be two or three pixels across the full-width at half-maximum (FWHM) of the PCLSF. This keeps the band-limited part of the spectroscopic signal well-sampled in the data. In general, interpolation works much better on critically sampled data than it does on ill-sampled data.

The fact that we are not interpolating the data means that we don't require critical sampling in the raw data. And indeed, there are spectrographs that don't have critically sampled pixel scales (for example, *APOGEE*; Majewski et al. 2017). Thus we will not assume that the raw data are critically sampled, but we will produce the output combined spectrum on a grid that is critically sampled with respect to the instrument resolution. The input and output pixel grids do not need to coincide when the model is a continuous function.

Non-uniform fast Fourier transform — In the special case that the spectral model is a Fourier series, there are very useful and fast tools for interpolation, including especially the non-uniform fast Fourier transform (for example, Barnett et al. 2019). These tools make the spectral combination substantially faster than naive linear algebra estimates would suggest.

Time variability — When we average data (or build a static model of multi-epoch data), we are implicitly assuming that we do not care about any of the intrinsic variability of the source being observed. That is, we are assuming that the changes from observation to observation are solely from either the shifts $\Delta\mathbf{x}_i$ or else from effects that we consider to be noise. We interpolate the model (not the data!) to account for the shifts, and we combine data to average down this noise. In many cases the stellar variability is itself the subject of study, and then the variability is not entirely noise. But from the perspective of this project, any stellar variability is considered to be part of the total additive noise in the spectrum. We return to this subject below in Section 5.

Bias, sky, tellurics, flat-field, normalization — In what follows, we will assume that the input data are well calibrated in various ways. In particular, we will assume that the images or spectra have been bias-corrected, sky-subtracted, flat-fielded, and tellurics-corrected sufficiently well that the different observations $\mathbf{y}_i$ can be seen as observations of the same underlying or intrinsic signal. Alternatively, the input data might be continuum-normalized**, pseudo-continuum-normalized, or flux-calibrated**. Again, the assumption is that the different calibrated or processed observations are of the same underlying signal. **No strict accuracy level is required in these pre-processing steps for *Frizzle* to be executed, but the consistency of the combined spectrum will naturally depend on the upstream processing steps.**

Many of these effects can be accounted for in straightforward generalizations of what we deliver below. For example, if there are multiple signals participating, the model would have to expand into something like *wobble* (Bedell et al. 2019). For another, if there are normalization variations, a normalization model can be added to the image combination procedure. **That is, if different input spectra have different normalizations, the model for the combined spectrum can be multiplied by a normalization model, unique to each input spectrum, prior to evaluation of the likelihood (e.g., Casey et al. 2026).**

Flux or photon conservation — Some methods for image combination, like `Drizzle` (Fruchter & Hook 2002), have as a hard constraint that they conserve *flux* or *photons*. We do not agree with this constraint or condition. The fundamental observable in a telescope**'s detector** is intensity, or energy per area per solid angle per time per wavelength (or per frequency). This quantity is related to the phase-space density of photons. When the mapping between a pixel grid and the sky is changed, the expectation of the rate of photon arrivals per pixel will change (because the pixel effective solid angles change), but the expectation of the intensity will not change. **Our position is that the spectra or images being combined make measurements of the same intensity at the detector surface; the combined image should provide a good mean estimate of this intensity.**

3. METHOD: *Frizzle*

The problem set-up is that there are N spectral measurements $\mathbf{y}_i$ (with $1 \leq i \leq N$), each of which is a M_i-pixel list of intensities (fluxes maybe)**, with scalar components $y_{i,k}$ for $1 \leq k \leq M_i$**. Different observations i might have different numbers M_i of pixels. The M_i pixels are associated with an M_i-length list of positions (wavelengths or log-wavelengths) $\mathbf{x}_i$ **with components $x_{i,k}$**. Each measurement or observation i is shifted by a shift operator $\Delta\mathbf{x}_i$, which can be thought of as a length-M_i list of pixel offsets or shifts **with components $\Delta x_{i,k}$**, such that the "rest frame" or "unshifted" positions corresponding to the length-M_i list of data $\mathbf{y}_i$ **are the differences $x_{i,k} - \Delta x_{i,k}$**. The goal is to get a combined spectrum that represents the mean of these spectra, in the common rest frame.

The spectral expectation model $f(x;\theta)$, which will be a representation of our combined spectrum, is a function of x controlled by a list θ of P linear parameters a_j (with $1 \leq j \leq P$). This function will be very flexible; possibly even an interpolation of control points or a Fourier series; the number P will be roughly the number of pixels we want in our output combined spectrum. The statistical model underlying can be expressed qualitatively as

$$y_{i,k} = f(x_{i,k} - \Delta x_{i,k}; \theta) + \varepsilon_{i,k} \tag{1}$$

$$f(x;\theta) = \sum_{j=1}^{P} a_j \, g_j(x) \ , \tag{2}$$

where $\varepsilon_{i,k}$ **is the additive (but not yet specified) noise contribution to pixel k of epoch i**, and each of the P functions $g_j(x)$ is a basis function of the representation.

The noise model is additive (as noted above) and we will assume that the noise contributing to observation $\mathbf{y}_i$ is zero-mean and Gaussian, with a known $M_i \times M_i$ variance tensor C_i (or its inverse C_i^{-1}). If the raw pixel measurements are independent, which they often are (and are often assumed to be), then the C_i (or C_i^{-1}) matrices will be diagonal matrices with the individual pixel variances (or inverse variances) down the diagonals. In many cases you don't know (or aren't provided with) the individual pixel variances, and you are just co-adding data with uniform weights. That choice, for our purposes, is equivalent to setting every inverse variance matrix C_i^{-1} to the $M_i \times M_i$ identity matrix. That's permitted! It is no worse a choice than it was (in the past) when you averaged your input data without thinking about noise variances or weights. Another sensible default choice **for photon noise-limited observations** would be to set each inverse matrix C_i^{-1} to the identity times the exposure time, such that the eventual math we do will weight the data by the exposure times. That is, you do not need to know the noise variances accurately (or even at all) to execute the forthcoming procedure; but if you do know them, it is good to use them.

If you additionally have a bad-pixel mask $\mathbf{b}_i$ associated with each observation $\mathbf{y}_i$, the bad-pixel mask can be used simply to zero out the corresponding bad rows and columns of the inverse covariance matrix C_i^{-1}. Importantly, the bad data get zeros in the *inverse* covariance matrix, not the covariance matrix!

Because we are often dealing with a spectrograph with a (fairly) well defined spectral resolution ($\delta\lambda/\lambda$ roughly constant), and because we care a lot about Doppler shifts (which are uniform in log wavelength), it makes sense to think of the pixel positions $\mathbf{x}_i$ and shifts $\Delta\mathbf{x}_i$ as being in log-wavelength (or ln-wavelength) units. Indeed, the *Sloan Digital Sky Survey* family of spectrographs extract one-dimensional spectra on a uniform-in-log-wavelength basis. Once that choice is made, there are still many choices for the linear basis, which include canonical functions from interpolation bases, wavelets, and Fourier series. In what follows, somewhat arbitrarily, we will use a Fourier basis, which can be expressed as follows:

$$g_j(x) = \begin{cases} \cos\left(\frac{\pi\,[j-1]}{L}\,x\right) & \text{for } j \text{ odd} \\ \sin\left(\frac{\pi\,j}{L}\,x\right) & \text{for } j \text{ even} \end{cases} , \tag{3}$$

where L is a (long) length-scale in the x space (the log-wavelength space), and $1 \leq j \leq P$.

Now for the method: We append all of the observations $\mathbf{y}_i$ into one huge M-pixel list Y, where $M = \sum_{i=1}^{N} M_i$. We make a $M \times M$ block-diagonal matrix C^{-1} from all the individual inverse variance matrices C_i^{-1}. We make a $M \times P$ design matrix X which is the evaluation of all the P basis functions at all the **rest frame** pixel locations $x_{i,k} - \Delta x_{i,k}$. Once we have these things, the best-fit parameters $\hat{\theta}$ are

$$\hat{\theta} = (X^\top\,C^{-1}\,X)^{-1}\,X^\top\,C^{-1}\,Y \;, \tag{4}$$

which is the maximum-likelihood or minimum-χ^2 solution for a linear weighted least

squares problem or a linear model with Gaussian noise (see, for example, Hogg et al. 2010). **Fitting a Fourier basis to non-uniformly sampled data by least-squares has a long history in least-squares spectral analysis (Vaníček 1971; Craymer 1998) and the closely related Lomb–Scargle periodogram (Lomb 1976; Scargle 1982). What is specific to our setting is the multi-epoch, shift-operator framing and the goal of returning a combined spectrum on a pixel grid rather than a periodogram.** And now if we choose a final (uniform, say) output pixel grid $x_\star$ in the rest-frame (unshifted) wavelength space, the combined spectrum is just

$$y_\star = X_\star \, \hat{\theta} \ , \tag{5}$$

where $X_\star$ is the $M_\star \times P$ evaluation of all of the P basis functions at the locations of the $M_\star$ pixels in the output combined-spectrum pixel grid $x_\star$. Putting these things together, the output spectrum $y_\star$ can be thought of as just a kind of linear weighted averaging operation on the input spectra $\mathbf{y}_i$ (which have been packed into Y) according to

$$y_\star = A_\star \, Y \tag{6}$$

$$A_\star \equiv X_\star \, (X^\top \, C^{-1} \, X)^{-1} \, X^\top \, C^{-1} \ . \tag{7}$$

That is, the method we are delivering is a linear combination of the input data, just like traditional interpolate-and-average. But our method never involves, not even implicitly, shifting or interpolating the input data.

The uncertainty on the combined spectrum $y_\star$ can be estimated in a few ways. The information-theoretic answer is that the **covariance matrix** $C_\star$ delivering the uncertainties on $y_\star$ is

$$C_\star = X_\star \, (X^\top \, C^{-1} \, X)^{-1} \, X_\star^\top \ . \tag{8}$$

The naive uncertainty variances are the **inverse** diagonals of this matrix. But, *very importantly*, this is only a correct usage when the input C_i matrices really represented the true uncertainties going in. If you have any concerns about that, then it is way safer to get your uncertainty estimates from bootstrap or jackknife (for a related example, see Hogg & Villar 2021).

In the end, most users want the combined spectrum $y_\star$ (and perhaps the pixel positions $x_\star$, uncertainty information, and propagated meta data). Some users will additionally want (or want to reconstruct) the parameters $\hat{\theta}$ and the basis functions $g_j(x)$. For these reasons it often makes sense to set $M_\star \geq P$ such that the delivered $y_\star$ fully determines the parameters. But that's a detailed question about your users and your goals.

Implementation notes — Although the equations like (4) show matrix multiplies between C^{-1} and X, it is not a good idea to perform these multplies as true matrix multiplies if the C^{-1} matrix is diagonal (as it often is). When C^{-1} is diagonal, the matrix multiply can be done as a broadcast of the vector of diagonal values instead, which is faster (and way less memory intensive).

And although the equations like (4) show inverses, it is almost never a good idea to use the native linear-algebra inverse function `inv()` in any implementation of this method. The reason is that we don't want the inverse matrix itself, we want the outcome of the inverse matrix multiplied by $X^\top C^{-1} Y$. For this use case, in `Python` it is more precise and more numerically stable to use the linear algebra `solve()` or `lstsq()` functions**, or comparable functions in other languages**. The only use for `inv()` is where the full inverse of a matrix is required, as when estimating the full $C_\star$ by (8).

Whenever performing `inv()`, `solve()`, or `lstsq()` operations, it is sensible to begin by checking the condition number (the ratio of the largest to the smallest singular value) of the matrix in play. Condition numbers larger than 10^8 are usually trouble, even when the mathematics is being performed at double precision. When condition numbers get large, maybe a different basis or a re-scale of some basis functions might be a good idea.

The condition number of the matrix $(X^\top C^{-1} X)$ will increase when multiple observations sample nearly identical rest-frame wavelengths $x_{i,k} - \Delta x_{i,k}$. This degeneracy can occur when two exposures happen to place pixels at very similar rest-frame positions, or in the pathological case of combining spectra that have already been resampled onto a common wavelength grid, which could render $(X^\top C^{-1} X)$ nearly singular. Provided we maintain $P < M$ (where $M = \sum_i M_i$ is the total number of input pixels) and employ robust `lstsq()` solvers, such near-degeneracies pose no computational difficulties in practice.

If X is either very sparse (as it is in a b-spline basis) or Fourier (as it is in a sine-and-cosine basis) then there are very fast ways to do the linear algebra. **For example, the implementation associated with this work uses the `finufft` package (Barnett et al. 2019) because we have a Fourier basis, and `JAX` (Bradbury et al. 2018) for automatic differentiation and just-in-time compilation, via the `jax-finufft` bindings (Foreman-Mackey & Garrison 2021).**

If you use any non-uniform fast Fourier transform package, the programming interface of that package will constrain how the frequencies must be ordered in the code, and it will probably require complex-number implementation. These have impacts on how the code is written. In particular the fact that the output combined spectrum $y_\star$ is real-valued constrains the complex Fourier amplitudes. Continuing on the Fourier theme: Fourier series, technically, model *periodic* functions. The length scale L of that periodicity in (3) must be set to a length longer than the x-space span of any of the input or output pixel grids.

We recommended setting $P = M_\star$. We recommend very strongly *against* setting $P \approx M$, where $M = \sum_i M_i$. That is, you want the number of parameters to equal your number of output pixels, but you *don't* want your number of parameters to be close to the number of *input* pixels. The reason for this is **the peak of the "double descent" curve at $P \approx M$, which de-stabilizes linear fitting** (Hastie et al. 2022). **In practice if** you find yourself in the situation that $M_\star \approx M$, reconsider your life choices.

4. EXPERIMENTS AND RESULTS

Fake data — We demonstrate the effectiveness of *Frizzle*—the spectral combination method by forward modeling—on artificial data. The spectral data were generated from a mean spectral expectation that is a continuum plus a randomly generated line list, where each line equivalent width was set by a random draw from a power law. **We arbitrarily assumed a mean density (Poisson rate) of 2×10^4 spectral lines per unit in x (natural logarithm of wavelength).** This line formation model is unphysical, but it is sufficient for our purposes in that it generates fake data that look like stellar spectra. Each spectral epoch was given a velocity (Doppler shift) by putting the epochs equally spaced on a sinusoidal variation with a semi-amplitude of $30\,\mathrm{km\,s^{-1}}$ (to emulate the motion of the spectrograph with respect to the Solar System barycenter). **The toy data set only constructed once, without adjustment: no tuning or selection was performed in order to make the data more or less favorable to a given method. Because the shifts in this demonstration are pure Doppler shifts, $\Delta x_{i,k}$ is constant in k for each epoch i.** The spectral expectation model is built on the assumption of a normalized (unit) continuum and unresolved spectral lines with a Gaussian line-spread function (LSF) with one-sigma width $1/R$, where $R = 135\,000$ is the spectrograph resolution.

The sampled data at each epoch were made by simply evaluating the spectral expectation model (shifted by the Doppler shift) at the pixel centers, such that the $R = 135\,000$ LSF is effectively the pixel-convolved LSF. After sampling the mean spectral expectation model onto an observational pixel grid, Gaussian noise is added with a variance that grows linearly with expected flux, such that the signal-to-noise per pixel in the continuum has a definite known value in each epoch spectrum. At a rate of 0.01 (one percent), sets of three adjacent pixels were randomly and independently marked as "bad" and offset by a large positive offset.

According to these rules we made two input data sets. One is poorly sampled, with $M = 171$ spectral pixels in the input data separated by $2/R$ in x (natural logarithm of wavelength), and signal-to-noise per pixel of 18 in the continuum. The other is well sampled, with $M = 340$ spectral pixels in the input data separated by $1/R$ in x, and signal-to-noise per pixel of 12 in the continuum. In each case **the noise is heteroskedastic, and** there are $N = 8$ epochs. The two complete data sets are shown in Figure 1 and Figure 2.

Frizzle — As our main experiment or result, we apply the forward model described above in Section 3 to the input data shown in Figure 1 and Figure 2. In each case, the output pixel grid $x_\star$ is chosen to be well sampled, with a pixel spacing of $1/R$ in the (natural) logarithm of wavelength. In order to give the model maximum flexibility, the number P of Fourier modes in the spectral model is set to be equal to the number of pixels $M_\star$ in the pixel representation. **In our experiments here, $M_\star = 338$.** The log-wavelength scale L of the Fourier basis was set to the pixel spacing times the number of pixels. **We were judicious in the choice of which pixels to include, as *Frizzle* is not robust to outlier pixels, which motivates a conservative approach to bad-pixel masking. This is no different to many other data analysis**

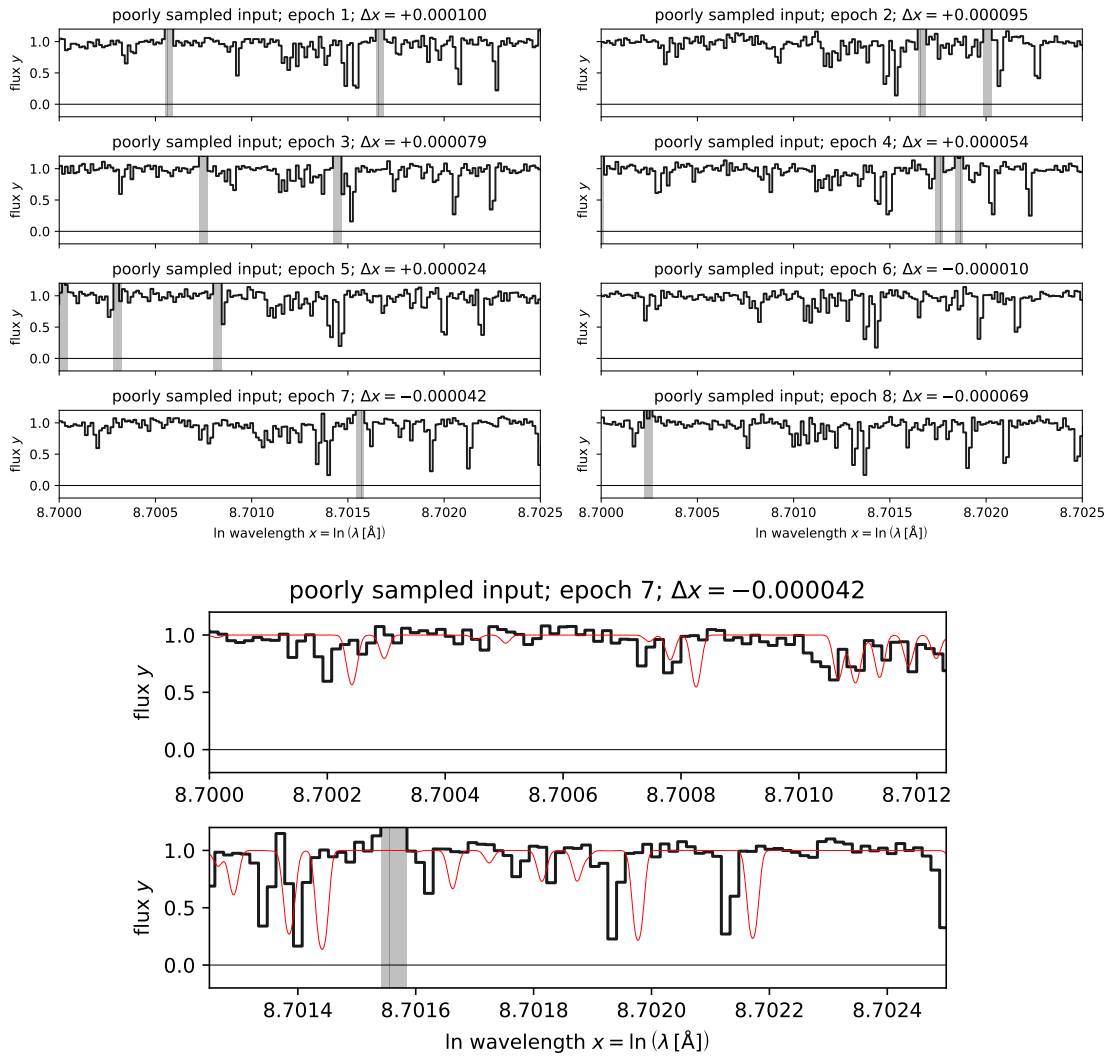

Figure 1. The top 8 panels show the $N = 8$ epochs of the poorly sampled input data, with $N = 8$, $M = 171$, and signal-to-noise per pixel of 18 in the continuum. The bad pixels are indicated with grey bars. The bottom 2 panels show a zoom in on one of the epochs, along with the true spectrum used to generate the data. The true spectrum is shown at zero Doppler shift, while each epoch spectrum is at a finite Doppler shift $\Delta \mathbf{x}_i$ relative to the spectrograph pixel grid.

situations in astrophysics, in that outliers (measurements not drawn from your model) will have a deliterious effect on your results. One approach to this would be to shift spectra to rest-frame, interpolate, and perform sigma-clipping to identify outliers. An alternative would be to use a robust regression method, such as iteratively re-weighted least squares (Holland & Welsch 1977), to downweight outliers in the forward modeling procedure. We leave these extensions to future work. Witout proper care, a 5-σ outlier value in the input data would have a 1-σ effect on the output spectrum if there are fewer than something like 25 epochs.

The combined spectrum resulting from the forward model described in Section 3 is shown in Figure 3. It is compared to the true spectral expectation model that was

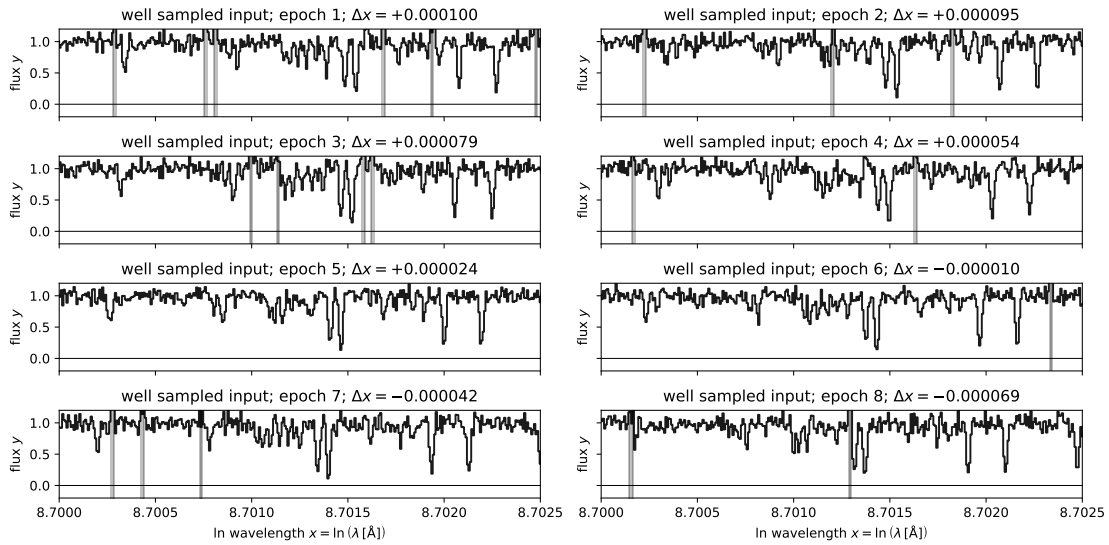

Figure 2. Similar to the top panels of Figure 1 except showing the well sampled input data with $N = 8$, $M = 340$, and signal-to-noise per pixel of 12 in the continuum.

used to generate the data. Also shown in Figure 3 is the residuals, *Frizzle* minus truth. The residuals look stationary and uncorrelated (more on this below).

Standard Practice(tm) — In standard practice, the individual spectra $\mathbf{y}_i$ are interpolated to rest-frame spectra $\mathbf{y}'_i$ using an interpolator. In detail, the data are shifted and also resampled onto the output pixel grid $x_\star$. The interpolator is a choice; cubic spline, Lanczos, and sinc interpolations are common, since they have good properties with respect to the spectrograph band limit. Here we choose a cubic spline for convenience, but there are many other choices, including linear, sinc, and Lanczos interpolators, among many others.

The interpolated spectra $\mathbf{y}'_i$ are, by construction, all on the same rest-frame wavelength grid $x_\star$ so they can be averaged, with a straight average, a median, an inverse-variance-weighted mean, or any more sophisticated algorithm. Here we choose **an inverse-weighted mean**, where we zero out the contributions of any bad pixels in that mean.

In Standard Practice(tm) **and in *Frizzle***, this censorship of the bad pixels requires some attention: Naive interpolation of an integer (binary actually) bad-pixel mask $\mathbf{b}_i$ will not deliver an integer mask $\mathbf{b}'_i$. The conservative move is to naively interpolate $\mathbf{b}_i$ (e.g., by a cubic spline) and then zero out any pixels that are significantly below unity. In general this substantially grows the binary mask, and reduces the amount of data available for combination. The non-conservative approach is just as bad: if we restrict the number (or fraction) of pixels in the combined spectrum to equal those in the individual epochs, we risk **using** genuinely bad pixels while growing the binary mask elsewhere and **marking good output pixels as bad**!

An example of a poorly-sampled spectrum interpolated to the rest frame is shown in Figure 4. The interpolation introduces substantial ringing artifacts in the spectrum (compare to Figure 1). It also grows the bad-pixel mask around the bad pixel.

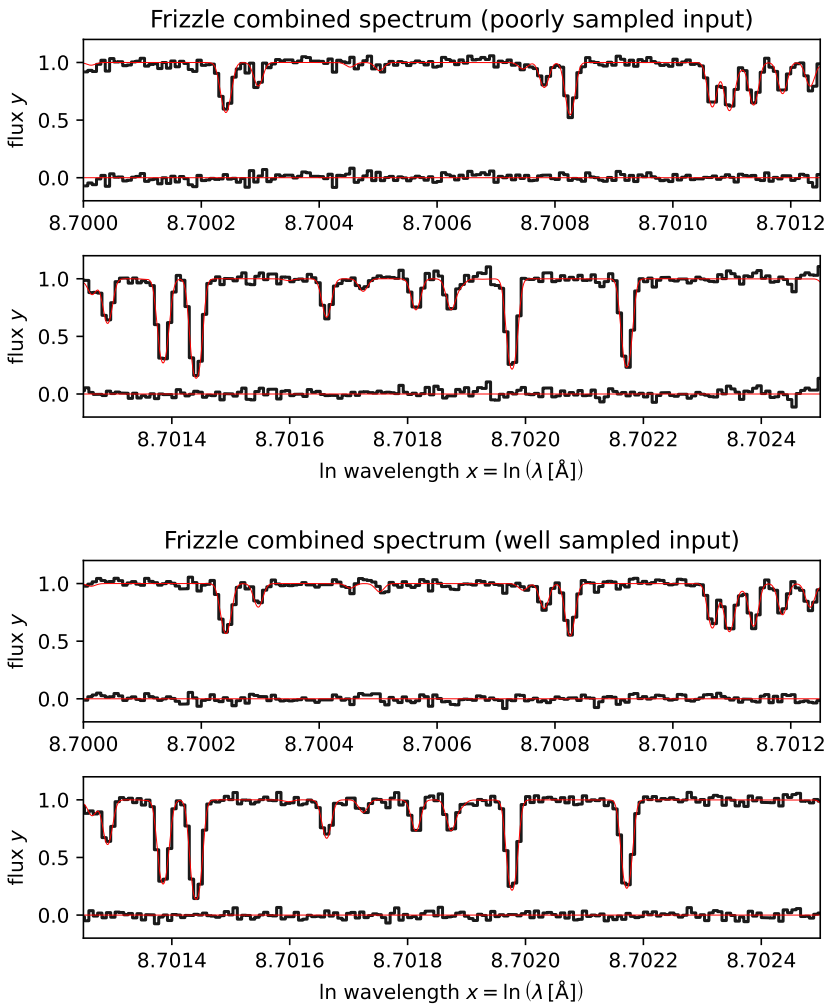

Figure 3. The top two panels show the result of running *Frizzle* on the poorly sampled input data, plus the residual away from the truth. The bottom two panels show the same but for the well sampled input data. In each panel, the red line shows the true model used to generate the data.

We show the final result of Standard Practice(tm)—interpolating the input data and then **simple averaging**—for the two data sets, in Figure 5. We also show the comparison with the true spectral expectation model that was employed to make the fake data. The residuals look correlated (there is ringing which mimics weak spectral features), and the residuals are larger near strong spectral features, especially in the case of the poorly sampled input data.

Noise and noise covariances — We repeated the above experiments to empirically measure the noise covariances, and to estimate how correlated neighboring pixels are in the output spectrum. For the poorly sampled case and the well sampled case, we generated 64 **sets of 8**-epoch data sets and computed the mean spectrum using *Frizzle* and Standard Practice(tm). We then computed the empirical covariance in the output spectrum minus the true spectrum, which are shown in Figure 6. In each case, Standard Practice(tm) has a lower noise variance at zero lag, but shows spatially

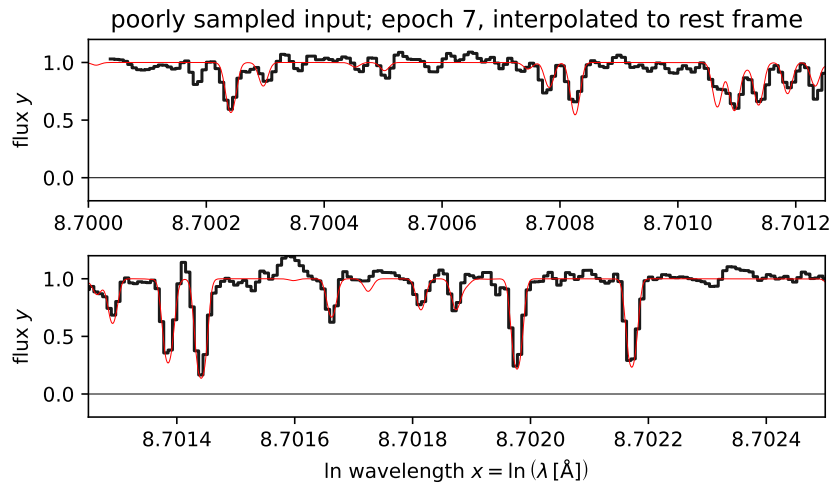

Figure 4. As part of Standard Practice(tm), it is necessary to interpolate the individual epoch spectra before averaging them. This shows one epoch of the poorly sampled data, interpolated to the output $x_\star$ pixel grid in the rest frame. The interpolated spectrum is compared to the true spectral expectation employed to generate it. Note the ringing induced by the interpolation. Also shown is the censored data, near an interpolated bad pixel and at the edge of the domain. Compare to the bottom panels of Figure 1.

correlated noise over many pixels. *Frizzle*, in contrast, shows no pixel-to-pixel noise covariances, even when the input data are ill sampled.

The scale of the induced correlation matches the interpolation stencil: a cubic spline uses four adjacent input pixels, so noise in one input pixel leaks into roughly four output pixels in the well-sampled case, and roughly eight output pixels in the poorly-sampled case (where the input spacing of $2/R$ is twice the output spacing of $1/R$), in agreement with the shapes of the Standard Practice(tm) curves in Figure 6. However, it is interesting to ask *why* the output *Frizzle* combined spectrum shows uncorrelated noise. It won't in general: If the number of epochs is small, or the coverage in the wavelength domain is spotty, there will in general be pixel-to-pixel covariances in the uncertainties on the combined spectrum **(e.g., particularly in the case of a single epoch spectrum that is under-sampled)**. However, if the number of epochs is large and the distribution of shift operators $\Delta\mathbf{x}_i$ is good, as it is in these experiments, the off-diagonal elements of the output combined-spectrum covariance matrix given in (8) will tend towards zero. The Standard Practice(tm) output combined spectrum always shows correlated noise, and especially when the input data are ill-sampled. The reason for this is that interpolation is very sensitive to noise in the ill-sampled case, such that the inter-pixel value of the interpolated function is moved substantially by small movements in the input pixel values, and these inter-pixel regions in the raw data are large relative to the pixel spacing in the combined spectrum spectral grid.

In a similar way, per-epoch continuum-normalization errors, which are smooth functions of wavelength by construction, can only contribute power at spatial frequencies well below the pixel-to-pixel scale, and so cannot in-

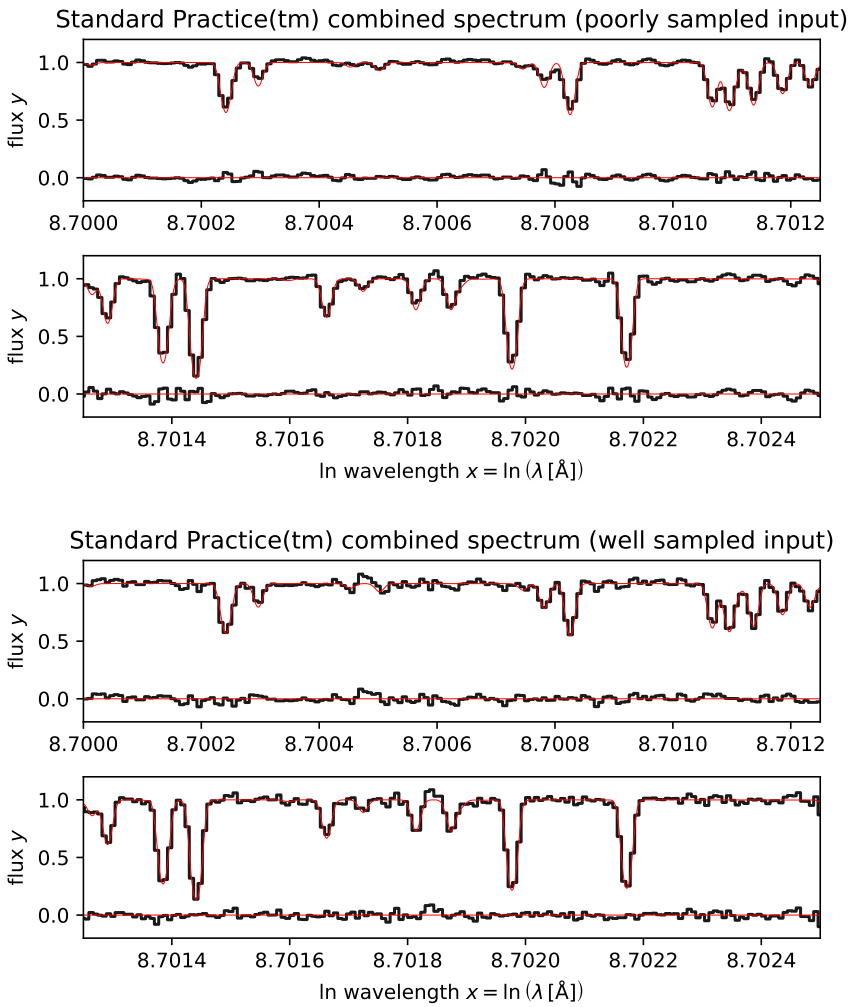

Figure 5. Same as Figure 3 but for the output of Standard Practice(tm). Note that in the poorly sampled case, the residuals appear to be spatially correlated, and the residuals appear to be larger near strong spectral features; more about these issues in the text and in Figure 6.

duce the kind of small-lag pixel covariances shown in Figure 6 for Standard Practice(tm); any such errors instead contribute to the variance at large lags and are common to both *Frizzle* and Standard Practice(tm). If the sampling in the input data are perfectly sampled, where one epoch is sampled precisely half a pixel offset from the first, and the next is precisely half pixel from the first, then in these situations standard interpolation will not suffer from correlated noise, because interpolation methods are specifically designed for regular sampling patterns. However, such situations are rare: it would require a stable spectrograph to observe a star with a stable relative velocity that precisely compensated for the difference in barycentric motion between two (and only two) epochs.

5. DISCUSSION

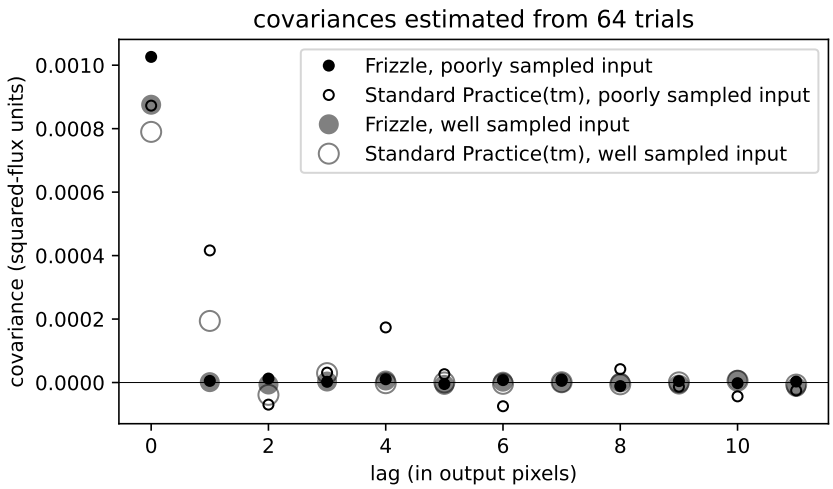

Figure 6. The empirical noise variance in the output (combined) spectra, and pixel-to-pixel covariances, as a function of pixel lag (pixel offset). The variances were estimated by performing the experiments repeated 64 times, with the same spectral expectations and sampling but unique noise draws. *Frizzle* has slightly larger variance per pixel at zero lag, but vanishing pixel-to-pixel correlations. In contrast, the Standard Practice(tm) leads to correlated pixel values, with correlations extending out to many pixels.

We have presented a forward-modeling approach to combining spectra or images; we name it *Frizzle*. *Frizzle* has good properties: It doesn't involve interpolating or otherwise distorting the input data; it produces a combined or mean spectrum in the format that astronomers traditionally expect; it makes no assumptions about the form of the combined spectrum (except that it can be represented on an $M_\star$-pixel grid); and it has excellent noise properties in real circumstances. The method can work on a wide variety of different kinds of data, but it is especially valuable in projects where the individual input exposures are ill-sampled, or not as well sampled as one might like; our original motivation was the individual *APOGEE* spectrograph (Wilson et al. 2019) exposures. It is also good for situations in which the input data have heterogeneous bad pixels or data gaps.

The method does not involve interpolating the data. However, it does fit a very flexible model to the data, and there is a limit in which fitting very flexible models is equivalent to interpolation (see, eg, Hogg & Villar 2021). So is this method, implicitly, interpolating the data? Our answer is no. Interpolation of the data involves non-parametric models with control points that are set by the pixel grids of the input data. Our model choice never refers to the properties of the input data pixel grids; it is chosen based on the desired properties of the output, combined-spectrum grid. Furthermore, interpolation functions (like cubic spline, say, or Lanczos-5, say) pass through every raw-data pixel value exactly. There is no point in this method at which the function being fit passes through every (or even any) of the raw-data pixel values.

In the Introduction (Section 1), we were agnostic about whether we were talking about imaging or spectroscopic data. In the end, our examples are spectroscopic. What differences are there between these cases? The main difference is that there

are many more "shift" parameters for imaging than spectroscopy: Two spectra (according to our assumptions) differ only (or primarily) in terms of a one-dimensional Doppler shift with respect to pixels that are located in one dimension (the wavelength direction). Imaging epochs, on the other hand, can differ in terms of two shifts and a rotation, or (equivalently) three Euler angles. And, since every pixel has its own two-dimensional position, the problem of locating pixels in an image is generally harder than the problem of locating pixels in an extracted spectrum. That is, the imaging case is generally more difficult. But it is not different, conceptually, from what is presented above. That is, once the (now more complicated) shift operators $\Delta\mathbf{x}_i$ are known, the problem proceeds by the same mathematics.

Everything presented above depends on a set of strong assumptions, listed in Section 2. Perhaps the strongest assumption is that the spectrum or image is not a function of time. That is, each epoch spectrum or image is a sampling of the same underlying, true signal. This assumption can be wrong for a myriad of reasons: The source can vary, the instrument can vary, there can be variable foregrounds or backgrounds, and there can be mistakes or variations in calibration or data processing. All of these things happen in real data sets, but *Frizzle* is no more impacted by this than Standard Practice(tm)!

There are (at least) three attitudes for an investigator to take in the face of variations in the data, from epoch to epoch or exposure to exposure: The first is that the procedures we design are designed to deliver the exposure (or signal-to-noise-squared) weighted mean of the observed epochs. That is, the fact that the data are varying does not mean that the methodology needs to be modified; without modification, the procedures given above—both *Frizzle* and Standard Practice(tm)—will produce something like the mean spectrum. With this attitude, the spectral variations are just another source of noise.

The second attitude to take—and the necessary attitude for many projects—is that the variations must be accounted for, by complexifying the forward model. For example, if there is a varying line-spread function between epochs (as there is for the *APOGEE* spectrograph, because the LSF is a function of the slit position of the optical fiber), then in addition to a shift at each epoch there is also an LSF adjustment at each epoch. The LSF adjustment is then applied to the spectral model before it is compared to each epoch of data, just as the shift is applied in the current methodological scope. Similar model adjustments can account for intrinsic spectral variations, additive or multiplicative nuisance signals in the data (like backgrounds or telluric absorptions), and mistakes in data pre-processing (like continuum modeling errors).

Although *conceptually* it is not hard to account for such variabilities, *in practice* they can make life hard. For one, the method we have presented assumes that we *know* the individual epoch shifts *a priori*, and all new epoch-to-epoch adjustments would also depend on per-epoch housekeeping data or parameters that might, in practice, be hard to know. That brings us to the third attitude that the investigator can take. What's presented above can be seen as a model for the *mean* of the process that generates individual-epoch spectra; that is the first term in an expansion of

moments. The next term would be the *variance* of the process. That is, a more sophisticated model could have not just P parameters for the mean spectrum, but additionally P' parameters for a variance model. If the variance is low dimensional, or sparse, or compact in important ways, there should be very effective models of this form. For example, we could imagine forward-modeling generalizations of some of sparse methods (for example, Candes & Romberg 2007) for applications in which the variability is localized in time or space; or we could imagine generalizations of matrix factorizations (for example, Tsalmantza & Hogg 2012) for applications in which variability is low dimensional (the *wobble* method has some such functionality; Bedell et al. 2019).

The forward-modeling step in *Frizzle* is fast in practice. Figure 7 compares the wall-clock cost of *Frizzle* against linear and cubic interpolation across a range of input sizes M; *Frizzle* is overhead-dominated at small M and scales comparably to cubic interpolation at large M, so computational cost need not be a barrier to adoption. Naively the cost of *Frizzle* is expected to be larger than its Standard Practice(tm) counterparts, but is accelerated by the use of non-uniform fast Fourier transform algorithms and just-in-time compilation. The overhead-dominated costs at small M are consistent with performance of non-uniform fast Fourier transforms. The code implementation that accompanies this work is written in `JAX`, allowing for execution on graphical processing units (GPUs). However, for the purpose of benchmarking, all timings were performed on a CPU.

The combined spectrum created by *Frizzle* has minimal pixel-to-pixel covariances based on repeated experiments. In contrast, the combined spectrum created by Standard Practice(tm) has substantial pixel-to-pixel covariances. This is particularly evident in the second panel of Figure 5 (say near $\ln \lambda = 8.7024$). Here the pixel-to-pixel covariances resulting from Standard Practice(tm) are not purely of academic interest, as the covariances here mimic weak spectral features. For example, if a weak Na line existed near this wavelength then a spectroscopist might readily interpret these covariances as real spectral features and report an enhanced Na abundance. Checking the strength of other (stronger) Na features would be a sensible check for this interpretation, but often there are few spectral lines available for a given element.

One side note about epoch-to-epoch variations in instrument or data-processing parameters: It is usually a good idea to model not the full instrument or data properties (as does Bolton & Schlegel 2010) if that can be avoided, but rather just the *differences* in instrument or data properties from one epoch to the next, taking those differences away from some fiducial data or instrument state. In the case of the LSF, for example, that obviates the need to model the infinite-resolution spectrum; the spectrum only needs to be modeled at fiducial resolution and convolved to other resolutions with difference kernels.

All the above considerations of variability bring us to another of our strong assumptions: We have assumed that we know the shifts $\Delta\mathbf{x}_i$ exactly. In practice, we rarely do, or maybe more accurately: We rarely know them as well as we could know them, in principle. When the shifts $\Delta\mathbf{x}_i$ are not known, or not known accu-

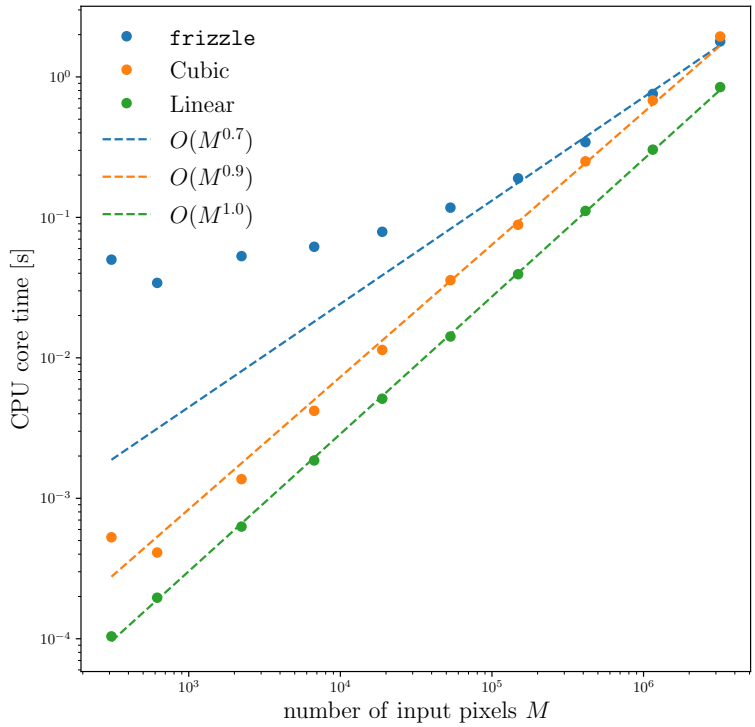

Figure 7. Computational cost (in CPU core seconds) for computing a combined spectrum and associated uncertainties with *Frizzle*, compared to linear and cubic interpolation. Linear and cubic interpolation scale linearly with the total number of input pixels M. *Frizzle* is overhead dominated at $M < 10^5$, and shows marginally sub-linear scaling at larger M, comparable to the cost of cubic interpolation. All timings were repeated, taking the best of repeated trials. *Frizzle* was executed on a CPU, but can be readily executed on graphical processors.

rately enough, we recommend a simultaneous-fitting procedure, in which the shifts are learned at the same time as the combined spectrum is estimated or inferred. This can be achieved through a challenging nonlinear optimization, or through an iteration of linearized optimizations, alternating improvements to the combined spectrum and improvements to the shifts. This is, more or less, how the *wobble* system works (Bedell et al. 2019).

We effectively assumed that the noise was Gaussian, and zero mean, but these are not always true of astronomical data. In principle even in the best-case scenarios there are Poisson noise sources, and cosmic rays and faulty detector pixels tend to add in highly skewed, non-zero-mean noise. The most important noise assumption made in all this is the zero-mean assumption; when this is wrong, all current spectral (or image) combination methods will be biased (including Standard Practice), unless they employ a likelihood function that accurately represents the detailed noise statistics.

We also assumed that there are $N > 1$ epochs, but in practice this is not always true. When there is only one epoch, *Frizzle* still has utility,

albeit limited. If there are no bad pixels then the number of modes $M_\star$ is equal to the number of input pixels M. In this case, the forward model is just a change of basis, and the output spectrum is just a re-representation of the input spectrum. If there are bad pixels, then the number of modes $M_\star$ is less than the number of input pixels M, which can cause the output spectrum to have high-frequency ringing. Usually these ringing artifacts are merely cosmetic, in that the covariance matrix $C_\star$ has large variances at the pixels affected by ringing, but such scenarios dance on the edge of numerical stability.

All code used for this project is available at https://github.com/davidwhogg/frizzle **and is registered in the Python Package Index as `frizzle`. This repository also includes code for generating the fake data sets, and for performing the experiments described above, as well as additional experiments thoughtfully requested during the review process.**

Software — `numpy` (Harris et al. 2020) — `scipy` (Virtanen et al. 2020) — `matplotlib` (Hunter 2007) — `pylops` (Ravasi & Vasconcelos 2020) — `finufft` (Barnett et al. 2019) — `jax-finufft` **(Foreman-Mackey & Garrison 2021)**.

It is a pleasure to thank **the anonymous referee for their detailed review, which helped us to improve this manuscript. We also thank** Mike Blanton (Carnegie Observatories), Matt Daunt (NYU), **Will Farr (Stony Brook/Flatiron)**, Dylan Green (UC Irvine), **Thomas Hilder (Monash)**, Jon Holtzman (NMSU), Dustin Lang (Perimeter), Adrian Price-Whelan (Flatiron), the Astronomical Data Group at the Flatiron Institute, and the Inference Group at Monash University for valuable discussions and input. The Flatiron Institute is a division of the Simons Foundation.

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